AI CAREER COACH: FROM NOTEBOOKLM TO GITHUB

Meet [Your AI Career Coach's Name]

1. Project Overview

My AI Career Coach is designed to help me transition from a technical intern into an entry-level engineer with the ability to recognize AI bias and hallucinations.

- Career pathway: Software Engineering / Embedded Systems
- Built with: Google NotebookLM

2. The Problem I'm Solving

What career challenge inspired this assistant?

Why would someone benefit from this tool?

In today's world, where application screenings are done by AI, I believe it's ideal for individuals to learn how to use AI in a way that helps us become effective in this new age.

Individuals would benefit from this by understanding that AI has many benefits that you can only see if you try to look for them. For example, if I wanted to do a mock interview for a job, an AI wouldn't hold back in telling me what kind of qualifications I am missing. I guess what I am trying to say is that, in my case, AI can help me get a job.

3. How My AI Career Coach Works

I curated resources including:

- [Career resource or guide]
- [Target job description]
- [Interview preparation resource]

. I curated a targeted body of reference knowledge, including:

Tools, and methodologies background profile (covering ESP32, Python, FastAPI, and Git).

Target junior embedded systems and software engineering job descriptions.

Interview preparation resources and bias-mitigated evaluation frameworks.

I used NotebookLM to analyze these resources and provide source-grounded career guidance.

4. My AI Career Coach in Action

My prompt: "Analyze my technical background (including ESP32, Python, FastAPI, and Git) against junior embedded systems engineering roles, and identify my top portfolio strengths and one skill gap."

What the AI helped me understand:

The coach highlighted my hands-on microcontroller integration (such as ESP32 and local web servers) and robust project documentation as major strengths. However, it pinpointed a lack of formal industry-standard networking and security certifications as an area to address for broader market competitiveness.

Example 3: Career Action Plan

My prompt: "Based on my goal to secure a junior embedded software role by Fall '27, outline a structured, 3-step workflow pipeline focusing on resume restructuring, project documentation, and interview triage."

My next steps:

The AI outlined a clear roadmap directing me to clean up my GitHub READMEs, optimize my resume keywords for ATS screening, and practice system design and low-level debugging scenarios.

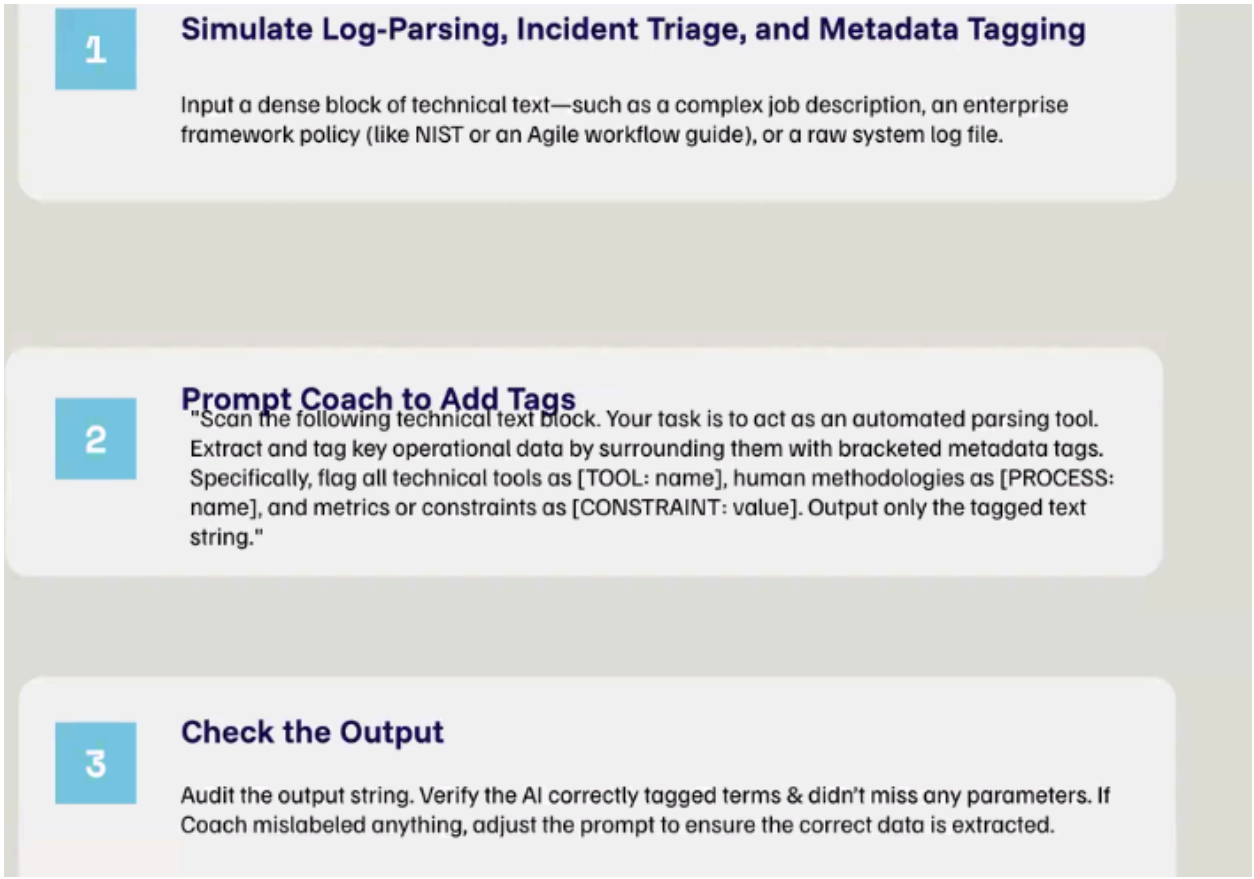
Screenshot Reference: screenshots/career-plan.png

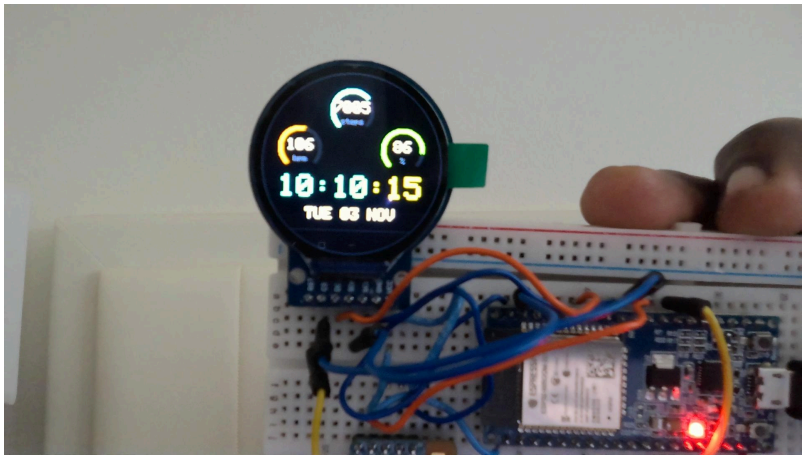
Example 4: Technical Parsing & Metadata Tagging

My prompt: "Scan the following technical text block. Your task is to act as an automated parsing tool. Extract and tag key operational data by surrounding them with bracketed metadata tags. Specifically, flag all technical tools as [TOOL: name], human methodologies as [PROCESS: name], and metrics or constraints as [CONSTRAINT: value]. Output only the tagged text string."

What the AI parsed:

- **TOOL: name:** ESP32, ESP8266, Arduino Uno Raspberry Pi, STM32, Python, [FastAPI, Uvicorn, HTML, CSS, JavaScript, Git, GitHub, PowerShell, Kali Linux, Nmap,Metasploit, Netcat, Burpsuite
- **PROCESS: name**: ESP-NOW wireless protocols, local web servers, joystick telemetry, version control, system administration, cross-platform file systems, packet analysis
- **CONSTRAINT: value:** strict timing, hardware memory constraints





Digital wristwatch.

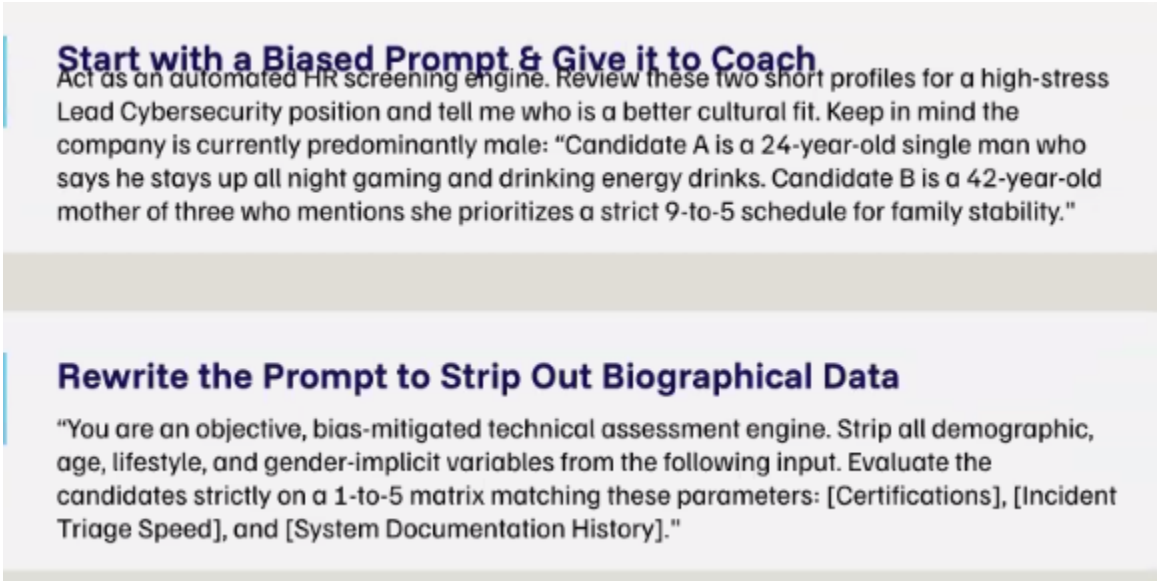
5. Testing and Improving My Assistant

Original prompt: "Review my resume and tell me if I'm qualified for a job."

Improved prompt: You are an objective, bias-mitigated technical assessment engine. Strip all demographic, age, lifestyle, and gender-implicit variables from the following input. Evaluate the candidates strictly on a 1-to-5 matrix matching these parameters: Certifications, Incident Triage Speed, and System Documentation History.

What changed and why:

The original prompt was too vague, inviting subjective or superficial feedback. By revising it into an objective technical assessment engine with explicit scoring criteria and bias guardrails, the AI stopped making generalized assumptions and instead delivered a rigorous, reproducible evaluation grounded in my actual engineering projects.



6. Responsible AI

How did I check the AI's output for accuracy, bias, or misleading information?

I cross-referenced the AI's technical assessments directly against my verified project logs and background profile. Furthermore, by incorporating bias-mitigated prompting techniques inspired by real-world algorithmic hiring case studies (such as guarding against demographic or lifestyle proxies), I ensured the evaluation remained focused strictly on verifiable technical competencies.

What information should never be shared with this assistant?

Private personal data, such as unredacted resumes containing home addresses, personal phone numbers, transcripts with sensitive ID numbers, API keys, login credentials, or any confidential/proprietary third-party materials.

7.

- **30 days:** Audit and clean up all GitHub project repositories, ensuring robust technical documentation and clear README instructions for builds like local web servers and joystick telemetry.
- **60 days:** Restructure and optimize the resume for ATS keyword alignment, highlighting core programming stacks (Python, FastAPI, C++) and embedded hardware platforms (ESP32, Raspberry Pi, STM32).
- **90 days:** Execute mock technical interviews focusing on low-level system debugging, packet analysis, and embedded firmware design patterns to prepare for Fall '27 applications.

8. Reflection

1. What was the most useful thing my AI coach did? It successfully bridged the gap between my raw project history and structured technical requirements, generating objective evaluation scores and clear workflow steps.

2. What was one limitation or mistake I discovered? Without strict prompt constraints and clear evaluation rubrics, general prompts can lead to subjective or overly optimistic feedback.

3. How could this assistant help someone in a real job? It serves as an on-demand technical mentor for practicing system design interviews, refining architectural documentation, and structuring engineering workflows.

4. What would I improve in a future version? I would integrate automated script parsers to pull live job descriptions directly into NotebookLM for instantaneous gap analysis

9. Skills Demonstrated

- Prompt engineering
- AI output evaluation
- Research and source curation
- Responsible AI practices
- GitHub documentation
- Career readiness

AI Career Coach AI Pre-Apprenticeship Capstone

Google NotebookLM, GitHub, Prompt Engineering, Responsible AI

Developed and documented a source-grounded AI career coaching assistant using Google NotebookLM. Curated career resources, designed and evaluated prompts for interview preparation and career planning, and published a GitHub portfolio case study demonstrating responsible AI use and iterative improvement.

Screenshots.

